

OSSP (Operating Systems And Systems Programming)

Course code: 25CS2104E

Report explaining how the OS abstracts CPU, memory, storage, and I/O devices and relationship between hardware resources and operating system services.

uname -a:

Displays information about the Linux operating system and kernel.

lscpu:

Displays information about the CPU, cores, threads, and architecture.

lsblk:

Displays connected storage devices and their partitions.

ps:

Displays currently running processes and their PIDs.

top:

Shows running processes and real-time CPU and memory usage.

CPU Abstraction:

The OS manages CPU time and schedules processes.

Memory Abstraction:

The OS allocates and manages memory for running processes.

Storage Abstraction:

The OS manages storage through files, directories, and file systems.

I/O Abstraction:

The OS manages I/O devices using device drivers.

Overall Abstraction:

The OS acts as a layer between applications and hardware.

Relationship: The OS manages hardware like CPU, memory, storage, and I/O devices and helps applications use them easily and safely.

